

PARAM PARMAR

Data Scientist · Machine Learning · AI Automation

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PROFESSIONAL SUMMARY

Aspiring Data Scientist and Computer Science undergraduate with strong foundations in Python, statistics, and machine learning. Experienced across the full data workflow — cleaning, exploratory analysis, feature engineering, model building, and evaluation — and comfortable translating results into clear, actionable insights. Pairs classical ML (regression, classification, and core deep-learning concepts) with modern generative-AI and automation tools such as LangChain, LangGraph, n8n, and MCP to build end-to-end, tool-connected solutions. Motivated to keep learning and to apply these skills to real-world data and AI problems.

CORE SKILLS & TOOLS

Python	Pandas	NumPy	scikit-learn	TensorFlow	Keras	Java
Matplotlib	Seaborn	OpenCV	LangChain	LangGraph	n8n	GitHub

- ★ **Machine Learning:** Supervised & unsupervised learning, regression, classification, model optimization, and core deep-learning concepts — CNN, LSTM, and Transformers.
- ★ **Generative AI & Automation:** LangChain, LangGraph, n8n, MCP, and AI APIs, with Pydantic for structured outputs — used to build tool-connected agents and multi-step automation workflows.
- ★ **Developer Tools:** GitHub, Visual Studio Code, Jupyter Notebook, Google Colab.

AI SKILLS & CERTIFICATIONS

- Certifications** — *Coursera · NPTEL*
- Python for Data Science — Coursera
 - Introduction to Machine Learning — Coursera
 - Database Management Systems & Data Mining — NPTEL

- AI Automation & Agent Workflows** — *Hands-on Practice*
- Built multi-step automation workflows in n8n, connecting AI APIs, webhooks, and external services to move and transform data across a pipeline.
 - Explored LLM application patterns with LangChain and LangGraph — prompt chaining, routing, and stateful agent graphs — using Pydantic and MCP for structured, tool-connected outputs.
 - Designed workflows with validation and fallbacks so model outputs stay structured, predictable, and easy to debug.

FEATURED PROJECTS (LEARNING-DRIVEN PORTFOLIO)

- Retail Sales Data Analysis** — *Python · Pandas · Seaborn · Matplotlib*
- Dataset / Context:** Public retail transactions dataset — 1,000+ records spanning product categories, customer demographics, and purchase dates.
- Cleaned and preprocessed 1,000+ retail transactions, resolving missing values, inconsistent formats, and duplicate records.
 - Engineered temporal and demographic features (month, weekday, age group) to enable time-based and segment-level analysis.
 - Analyzed revenue trends across product categories, customer demographics, and time periods using grouped aggregations.
 - Built visualizations with Matplotlib and Seaborn to communicate patterns, outliers, and category-level comparisons clearly.
 - Translated the analysis into concise, actionable business recommendations.
 - Key Learning: end-to-end EDA workflow — from raw-data cleaning and feature engineering to visualization and insight communication.

- House Price Prediction Using Linear Regression** — *Python · Pandas · scikit-learn*

- Dataset / Context:** Structured housing dataset used to model and predict property prices from numerical and categorical features.
- Built a linear regression model in scikit-learn to predict house prices from structured housing data.
 - Prepared the data through cleaning, feature engineering, and encoding, then split it into training and test sets.
 - Evaluated model performance using RMSE and R² to quantify accuracy and goodness of fit.
 - Interpreted coefficients and residuals to understand which features most influenced the predicted price.
 - Key Learning: the supervised ML pipeline — preprocessing, model training, and quantitative evaluation.

Adaptive RAG — Agentic Retrieval-Augmented Generation — Python · LangChain · LangGraph · MCP · LLM APIs

- Dataset / Context:** A document question-answering system that adapts its retrieval strategy to the complexity of each query.
- Built an agentic Retrieval-Augmented Generation pipeline in LangGraph that answers simple queries directly and routes complex ones through single- or multi-step retrieval.
 - **AI-Core Feature** — Implemented an LLM-based router with Pydantic-structured outputs that classifies query complexity and grades the relevance of retrieved documents, re-retrieving when results are weak.
 - Integrated a vector store for semantic document retrieval and connected external tools and services through MCP for structured, tool-aware responses.
 - Added self-correction and fallback logic: when retrieved context is insufficient, the graph refines the query or broadens retrieval before generating a final answer.
 - Structured the workflow as a stateful graph so each stage — route, retrieve, grade, generate — stays modular and debuggable.
 - Key Learning: designing agentic LLM workflows — query routing, retrieval grading, and structured, self-correcting generation.

EDUCATION

Bachelor of Engineering — Computer Science Engineering Sal Institute of Technology and Engineering Research (GTU)	Expected 2027
Class XII A One School (GSEB)	2023
Class X A One School (GSEB)	2021

Relevant Coursework: Database Management Systems, Data Mining (NPTEL); core Computer Science fundamentals.

ACHIEVEMENTS & EXTRACURRICULARS

- Strengthened core computer-science foundations through NPTEL coursework in Database Management Systems and Data Mining.
- Completed multiple self-driven data science and machine learning projects, taking each from raw-data cleaning through model evaluation and insight communication.
- Comfortable presenting data insights and AI project walkthroughs to non-technical audiences.